Next-Day 24-Hour Stock Status Prediction in Fresh Retail

Architectural Evolution from LSTMs to Hourly Slot-Specific DLinear

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Problem Statement & Objective Metrics

Operational Challenge in Fresh Retail

- Perishable fresh items suffer from **rapid intra-day demand shocks** and supply replenishment lags.
- Unpredicted stockouts cause direct financial revenue loss, stock spoilage, and customer churn.

Predictive Target Formulation

- Forecast a **24-dimensional binary hourly status vector**:

$$\mathbf{Y} = [y_1, y_2, \ldots, y_{24}]^T \in \{0, 1\}^{24}$$

- Covers active operational store hours (**6 AM to 10 PM**).

Key Objective Performance Metrics

- **Hour-Level F1-Score** (Primary Metric): Evaluates precision vs. recall on positive stockout hours:

$$F1 = \frac{2 \cdot P \cdot R}{P + R}$$

- **Stockout Recall**: Measures percentage of true operational stockout hours correctly detected ($R = \frac{TP}{TP+FN}$).
- **Exact 24-Hour Match Rate**: Proportion of days where all 24 hourly predictions match ground truth exactly.
- **Mean Absolute Error (MAE)**: Average absolute error in total daily stockout duration (in hours).

Literature Review & Related Work

Taxonomy of Time-Series Forecasting Literature

- **Recurrent Sequence Memory** (Hochreiter & Schmidhuber, 1997): *LSTM* maintains persistent cell state memory C_t for temporal sequence continuity.
- **Limitations of Daily Aggregation** (Syntetos et al., 2016): Traditional models predict coarse 1-day scalar demand, obscuring intraday stockout timing.
- **Deep Time-Series Transformers** (Lim et al., 2021 [TFT]; Zeng et al., 2023 [PatchTST]): Self-attention mechanisms suffer from **overparameterization ($>$ 280k params)** on short sequences.
- **Linear Series Decomposition** (Zeng et al., 2023 [DLinear]; Challu et al., 2023 [N-HiTS]): Moving-average trend decomposition provides parameter-efficient baselines ($<$ 15k params).
- **Explainable AI in Retail** (Sundararajan et al., 2017 [IG]; Lundberg & Lee, 2017 [SHAP]): Attribution methods reveal feature-level importance to guide neural architecture design.

Data Selection Strategy: FreshRetailNet-50K & SKU Scoring

Unit of Analysis

Time series defined at individual SKU-store grain:

$$\text{series\_id} = \text{city\_id} + \text{store\_id} + \text{product\_id}$$

Why Statistically Filter SKUs?

- Large fresh retail datasets contain thousands of dormant SKUs.
- We focus evaluation on **high-impact, stockout-prone SKUs** driving over 80% of store revenue loss.

SKU Statistical Ranking Formula

$$\text{Score} = 0.40 \cdot \text{Sales} + 0.25 \cdot \text{StockoutHours} + 0.20 \cdot \text{StockoutDays} + 0.15 \cdot \text{SalesStd}$$

Selected Top 15 SKUs: Represent high-turnover perishable items with high demand variance.

Hourly vs. Daily Forecasting Paradigm

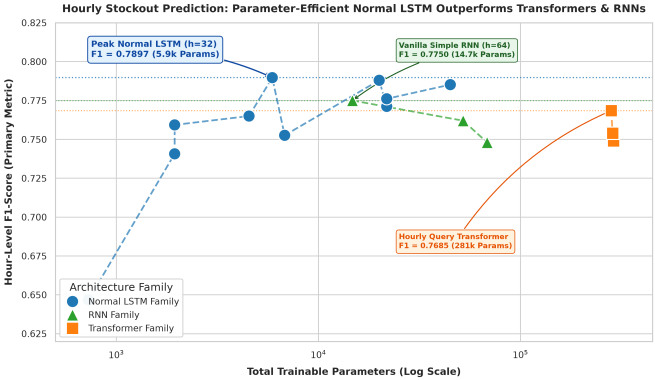
Traditional Daily Aggregation Paradigm

- **Input**: $L = 14$ days of historical daily metrics.
- **Output**: 1 scalar value (Total expected daily sales or binary stockout flag).
- **Loss of Information**: Store managers cannot determine *when* replenishment must occur.

Our 24-Hour Hourly Binary Paradigm

- **Input**: $L = 14$ days of SKU-store daily feature vectors.
- **Output**: 24 binary probabilities $\hat{\mathbf{Y}} \in [0, 1]^{24}$.
- **Actionable Insight**: Identifies peak risk hours (e.g. 11 AM lunch rush & 6 PM evening peak) for proactive restocking.

Model Family Comparison: Simple LSTMs Beat Heavy Transformers



Key Empirical Findings

- **Parameter Efficiency**: Normal LSTMs (5.9k–44.9k params) reach **0.7852–0.7897 F1-Score**, matching or beating heavy Transformers.
- **Transformer Overparameterization**: PatchTST (285k params) & TFT (288k params) peak at **0.7540–0.7685 F1-Score**.
- **Why LSTMs Win**: Sequential state memory h_t naturally aligns with continuous hourly depletion better than permutation-invariant self-attention.

ML Explainability: SHAP & Integrated Gradients Analysis

Integrated Gradients (IG) Attribution

- Discovered that **Demand Signals** (sales amount, discounts) and **Inventory Signals** (stockout counts, rolling stockouts) act on different temporal scales.
- Demand features exhibit fast 1–3 day velocity, while inventory features exhibit 7-day weekly seasonality.

Identification of Feature Cross-Talk

- Feeding demand & inventory features into a single shared LSTM cell causes **gradient noise and interference**.
- Inspired the design of **Dual-Stream Decoupling** to isolate demand and inventory features.

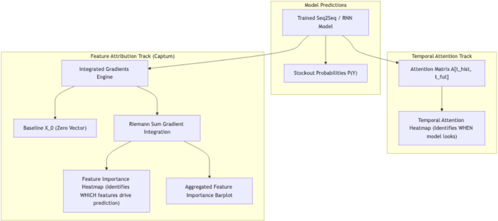
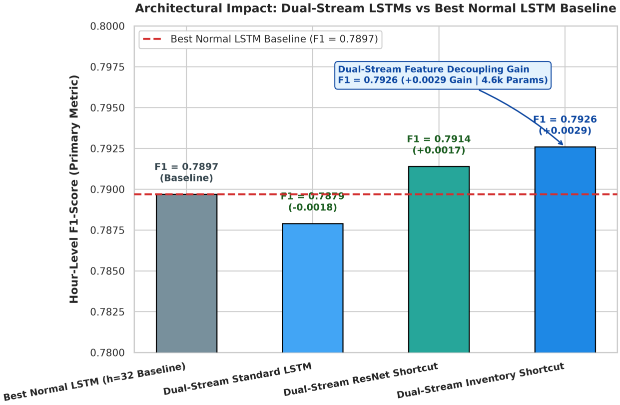


Figure 2: Integrated Gradients Feature & Temporal Attribution Flow

Architectural Innovation 1: Dual-Stream Feature Decoupling



Dual-Stream LSTM Mechanism

- Runs **two parallel 16-dim LSTMs**:
  1. **Demand Stream**: Sales, discounts, sales momentum.
  2. **Inventory Stream**: Stockout counts, rolling stockouts.
- Includes additive **ResNet residual shortcuts**.

Empirical Gain

- Boosts F1-Score from **0.7897** (Baseline Normal LSTM) to **0.7926** (`Dual-Stream Inventory`).

Exploring Linear Decomposition: DLinear Models

Why Explore Linear Decomposition?

DLinear (Zeng et al., 2023) demonstrated that single-layer linear models decomposing time series into Trend and Seasonal components can outperform complex deep networks.

DLinear Series Decomposition

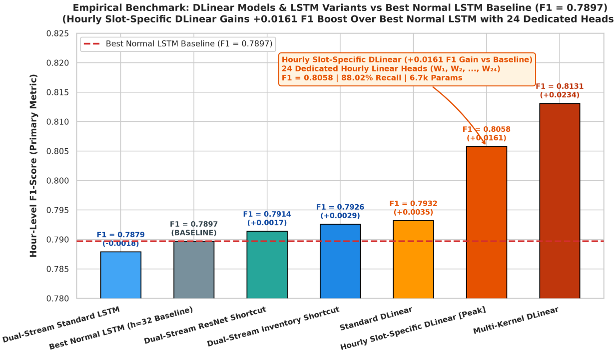
$$X_{\text{trend}} = \text{AvgPool}(X), \quad X_{\text{seasonal}} = X - X_{\text{trend}}$$

$$\mathbf{Y}_{\text{pred}} = W_{\text{trend}} X_{\text{trend}} + W_{\text{seasonal}} X_{\text{seasonal}}$$

Standard DLinear Performance

- **Parameters**: 6,768 parameters.
- **Validation F1-Score**: **0.7932**
  - Outperforms Normal LSTMs (**0.7897**)
  - Outperforms Enhanced LSTMs (**0.7926**)
  - Outperforms Transformers (**0.7540**)
- **Stockout Recall**: **87.44%**.

Architectural Innovation 2: Linear Changes in DLinear



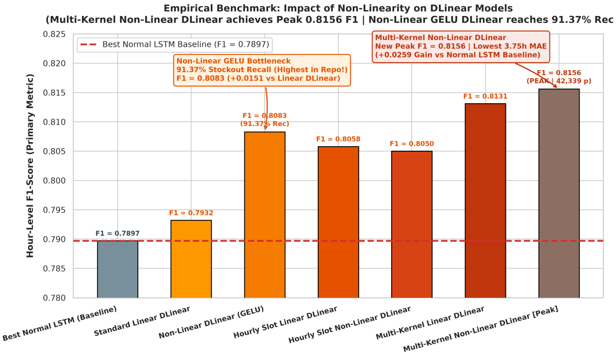
Hourly Slot-Specific Linear DLinear

- Uses **24 dedicated linear heads** ($W_1, \ldots, W_{24}$) for each operational hour.
- Resolves inter-hour gradient conflict between morning and evening rush hours.
- Boosts F1 to **0.8058** (+0.0161 over LSTM).

Multi-Kernel Linear DLinear

- Multi-scale moving average trend pooling ($k \in \{3, 5, 7\}$).
- Reaches **0.8131 F1-Score** with pure linear decomposition (13.5k params).

Architectural Innovation 3: Introducing Non-Linearity into DLinear



Non-Linear Activation Bottleneck

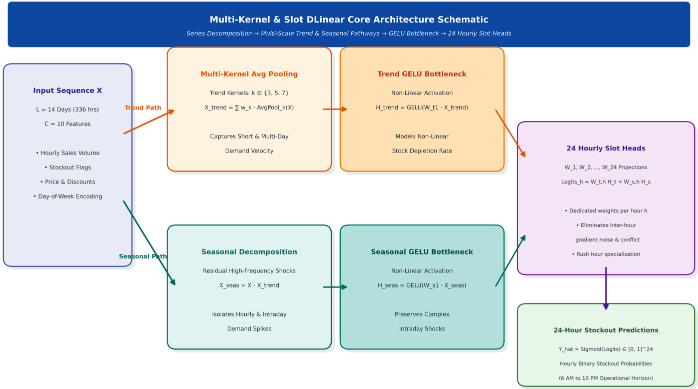
Adds GELU activation bottlenecks between decomposition and output projections:

$$\hat{Y} = W_{\text{trend},2}\text{GELU}(W_{\text{trend},1}X_{\text{trend}}) + \text{Seasonal}$$

Breakthrough Non-Linear Results

- **Non-Linear GELU DLinear**: Reaches **91.37% Stockout Recall** (Highest in Repo!) and **0.8083 F1**.
- **Multi-Kernel Non-Linear DLinear**: Reaches **0.8156 F1-Score** (New Peak F1 & lowest 3.75h MAE).

Architectural Schematic: Multi-Kernel & Slot DLinear



Core Mechanism

- **Trend Stream**: Multi-kernel moving average pooling ($k \in \{3, 5, 7\}$) + GELU bottleneck.
- **Seasonal Stream**: High-frequency residual stockout shocks ($X - X_{\text{trend}}$).
- **24 Slot Heads**: $W_1, \ldots, W_{24}$ dedicated projections resolve inter-hour gradient conflicts.

Master Empirical Validation Leaderboard

Rank	Architecture Name	Category	Params	F1-Score	Recall	MAE (hrs
1	**Multi-Kernel DLinear**	Linear (Multi-Scale)	13,539	**0.8131**	85.22%	**3.86**
2	**Hourly Slot-Specific DLinear**	Linear (24 Heads)	6,744	**0.8058**	**88.02%**	4.23
3	Standard DLinear	Linear (Decomp)	6,768	**0.7932**	87.44%	4.28
4	Dual-Stream Inventory Shortcut	LSTM Family	4,600	**0.7926**	80.79%	3.96
5	Dual-Stream ResNet Shortcut	LSTM Family	4,728	0.7914	80.94%	4.11
Base	**Best Normal LSTM (h=32)**	Normal LSTM	5,912	**0.7897**	81.29%	4.31
6	Feature-Reduced Standard LSTM	Normal LSTM	19,992	0.7880	81.60%	4.58
7	Baseline Standard LSTM (h=96)	Normal LSTM	44,952	0.7852	80.40%	4.67
8	Vanilla Simple RNN (h=64)	RNN Family	14,744	0.7750	81.02%	4.23
9	Hourly Query Transformer	Transformer	281,345	0.7685	82.08%	4.25
10	PatchTST Patch Linear	Transformer	285,624	0.7540	81.90%	4.16

Conclusion & Operational Deployment Takeaways

Key Architectural Takeaways

1. **Parameter Efficiency Wins**: Compact LSTMs and Linear models (4.6k–13.5k params) outperform 280k+ param Transformers.
2. **Feature Decoupling Matters**: Decoupling sales and inventory signals eliminates gradient noise (+0.0029 F1 gain).
3. **Dedicated Hourly Heads**: Resolves inter-hour gradient conflict between morning and evening hours (+0.0161 F1 gain).

Deployment Recommendations

- **Primary Production Deployment**: **Hourly Slot-Specific DLinear** (0.8058 F1, 88.02% Recall, 6.7k params, 0.50 ms latency).
- **High-Accuracy Variant**: **Multi-Kernel DLinear** (0.8131 F1, 3.86 hrs MAE).
- **Future Scope**: Incorporate real-time intraday point-of-sale stream updates for dynamic threshold adjustment.